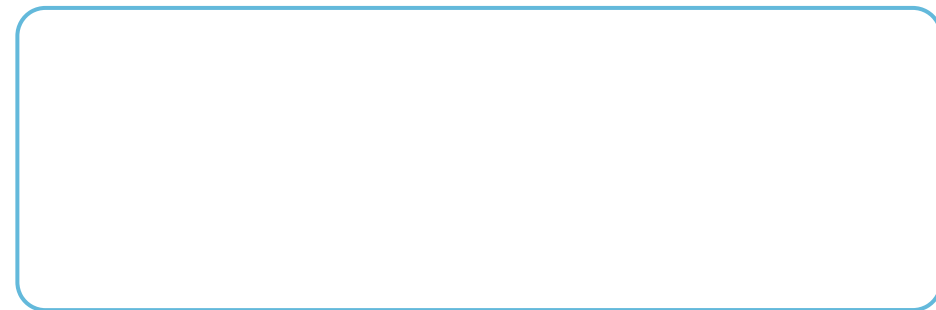


Python for data analysis

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Session's program

- 1) Essential tools to install
- 2) pandas library
- 3) xarray library
- 4) Model, in-situ and altimetry data

What's your OS?

Linux, Mac: all good, use terminal.

Windows: Let's install WSL2 to simulate Linux.

- Open a Power shell as administrators, and type:
 - `wsl --install -d Ubuntu`You can also install Ubuntu from the Microsoft store.
- Reboot your computer if asked so.
- Run Ubuntu from the start menu, and wait a moment.
- A **UNIX username** and **password** will be asked.
- From now on, enter all the commands in an Ubuntu terminal.
- To verify if you have ubuntu:
 - `uname -a` #Must mention 'Linux' and 'microsoft-standard-WSL2' on windows
 - `whoami` #You Linux username

Windows only: storing files

- Put your files in the Linux home directory (`~`), not under `/mnt/c/...`. Working from the Windows drive is many times slower. To copy the YAML and the course folders over from Windows (replace `<WinUser>` with your Windows user name):
 - `cp /mnt/c/Users/<WinUser>/Desktop/ggosss2026_conda_env.yaml ~/`
 - `cp -r /mnt/c/Users/<WinUser>/Documents/GGOSSS_2026/open_drift ~/`

To see the Linux files from Windows Explorer, go to `\\wsl$\\Ubuntu\\home\\<linux-user>`.

Conda

Conda installs and isolates python libraries in a closed environment to avoid conflicts with the rest of your system.

- Download and install Miniconda:
 - `cd ~`
 - `wget -c https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh`
 - `bash Miniconda3-latest-Linux-x86_64.sh`
- Accept the license, keep the default location (`~/miniconda`) and answer **yes** when you're asked to initialize.
- Close and reopen a terminal. Your command line should start by `(base)`, this is the name of your environment.
- Confirm by typing: `conda --version`

Create an environment

We want to create a dedicated environment for this school, where we'll install all the required python libraries. We just need to create the environment once. Then we can activate it every time we open a terminal.

- We'll install every package using the conda-forge channel. Taking Anaconda's `defaults` channel out of the configuration prevents mixed-channel conflicts.
 - `conda config --add channels conda-forge`
 - `conda config --remove channels defaults` #ignore "not found" if it isn't here
 - `conda config --set channel_priority strict`
- Clone the github repository and access it:
 - `git clone https://github.com/GGOSSS-Summer-School/GGOSSS-2026.git`
 - `cd GGOSSS-2026`
- Create the environment using the YML file:
 - `conda env create -f ggoss2026_conda_env.yml`
- To activate the environment:
 - `conda activate ggoss26`

You need to reactivate the environment each time you open a terminal!

Get the material of this session



Data:

- Download data from the drive:
https://drive.google.com/drive/folders/1a9_RhJpIXdrU36dJ1LmsJAXfxU0ij0F_?usp=sharing
- Or take it from a USB key

Put them in the `GGOSSS-2026/01-python-tutorial/data` directory

Notebooks:

- They should already be in `GGOSSS-2026/01-python-tutorial`
- Otherwise, use one of the two methods above.

Jupyter Notebook

Many of the sessions of the school are using Jupyter Notebooks. They combine code (Python) and text (markdown)

Jupyter is already installed in the environment we create.

Run jupyter lab in the working directory

- `cd GGOSSS-2026`
- `jupyter lab`

We will now go through the following notebooks

- `pandas.ipynb`
- `xarray.ipynb`
- `croco_insitu_satellite.ipynb`

Questions?

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